

Depth-Conditioned Feature Compositing for Unsupervised Monocular Panoptic Segmentation

From Strong Baseline to Structural Diagnosis: Elevation via Orthogonal Depth-Semantic Fusion

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Abstract

We establish that monocular unsupervised panoptic segmentation can surpass the stereo-dependent state of the art. A simple composition of frozen monocular depth estimation (Depth Anything v3) and self-supervised semantic clustering (CAUSE-TR, $k=80$) already generates pseudo-labels achieving 27.87% PQ on Cityscapes—matching CUPS (27.8%, CVPR 2025) which requires stereo video and optical flow, and exceeding it to 32.76% PQ after EMA self-training with DINOv3 ViT-B/16 + Cascade Mask R-CNN. Yet this strong baseline plateaus: depth-split instances over-fragment catastrophically (~ 44 per image, 50.7% stuff contamination), and semantic clusters from frozen DINOv3 lack geometric awareness, leaving a persistent 3–4 PQ gap to true panoptic coherence. We diagnose these as *structurally orthogonal* error modes—feature-level, geometric-level, and label-level—and show they compose near-additively by introducing two lightweight interventions: a **Depth-Conditioned Feature Adapter** (DCFA, 40K parameters) that modulates frozen DINOv3 features with sinusoidally-encoded depth, raising semantic clustering by **+2.60 mIoU**; and a **Semantic-Instance Mutual Consistency Filter** (SIMCF-ABC) that merges fragmented depth instances via DINOv3 boundary similarity, improving thing-class quality by **+1.48 PQ_{th}**. The combined +1.31 PQ pseudo-label gain amplifies non-linearly through EMA self-training to +4.21 PQ in the trained network, pushing the final result to **35.83% PQ**—**+8.0 PQ above CUPS**—with zero-shot transfer to Mapillary Vistas (39.19% PQ) and KITTI (34.85% PQ).

Keywords: unsupervised panoptic segmentation, monocular depth estimation, DINO, pseudo-label generation, self-training, cross-modal fusion, error decomposition

1. Introduction

Panoptic segmentation unifies semantic segmentation (assigning each pixel a class label) with instance segmentation (distinguishing individual objects), producing a complete scene understanding where every pixel belongs to either a countable “thing” instance or an amorphous “stuff” region [1]. Supervised panoptic models have achieved remarkable accuracy, but their dependence on dense per-pixel annotations—which cost approximately 1.5 hours per image on Cityscapes [2]—severely limits scalability to new domains and categories.

From stereo video to monocular images. The state of the art in unsupervised panoptic segmentation, CUPS [3], achieves 27.8% PQ on Cityscapes by leveraging stereo video sequences and optical flow for multi-view consistency. This dependence on temporal geometry fundamentally restricts applicability to monocular image collections—the dominant modality in most real-world scenarios. We ask: *can a fully monocular pipeline, using only single RGB images, not only match but exceed the stereo-dependent state of the art?*

The answer, surprisingly, is yes—even before introducing any novel architectural components. A simple composition of two frozen foundation models—monocular depth estimation (Depth Anything v3 [13]) for geometric boundaries, and CAUSE-TR [9] with $k=80$ overclustering for semantic categories—generates panoptic pseudo-labels that already achieve 27.87% PQ

on Cityscapes, matching CUPS’s 27.8% PQ while using strictly less input data. Training a Cascade Mask R-CNN with frozen DINOv3 ViT-B/16 backbone via EMA self-training on these pseudo-labels pushes the result further to 32.76% PQ, establishing a strong monocular baseline that surpasses the stereo-dependent SOTA by +5.0 PQ.

The plateau and its diagnosis. Yet this strong baseline hits a ceiling. Depth-guided instance splitting over-fragments catastrophically: each image yields ~ 44 instances with 50.7% stuff contamination, because depth discontinuities alone cannot distinguish between a single large object (one instance) and adjacent surfaces (background stuff). Meanwhile, semantic clustering from frozen DINOv3 features lacks geometric awareness, assigning the same label to a car hood and the road beneath it when their appearance features are superficially similar. The fusion of these two imperfect streams introduces cross-modal inconsistency: semantic regions over-spill instance boundaries, and depth fragments split semantically coherent objects.

We diagnose these failures as *three structurally orthogonal error modes*:

- **Feature-level errors.** Frozen self-supervised features encode appearance but are blind to geometry. A lightweight depth-conditioned projection (40K parameters) that modulates DINOv3 patch features with sinusoidally-encoded monocular depth improves semantic clustering mIoU from 52.69% to

55.29% (+2.60).

- **Geometric-level errors.** Monocular depth provides natural instance boundaries, but raw splitting produces catastrophic over-fragmentation. Alternative post-processing algorithms (Canny, watershed, multiscale Sobel) provide *zero improvement* over simple Sobel+CC—the binding constraint is depth estimator quality, not splitting sophistication.
- **Label-level errors.** Merged semantic clusters and depth instances suffer cross-modal inconsistency. A three-step consistency filter—semantic majority voting, feature-guided instance merging via DINOv3 cosine similarity, and depth outlier masking—resolves these conflicts and contributes +1.48 PQ_{th} alone.

The punchline: near-additive composition. These three interventions target structurally independent error modes and compose near-additively: DCFA (+0.68 PQ) and SIMCF-ABC (+0.73 PQ) sum to +1.31 PQ (93% additive). This +1.31 gain in pseudo-label quality amplifies non-linearly through EMA self-training to +4.21 PQ in the trained student network, reaching **35.83% PQ** on Cityscapes validation—**+8.0 PQ above CUPS** under the same 27-class CAUSE+Hungarian protocol, using only monocular RGB at inference.

Contributions. Our contributions are:

- **A monocular pipeline that surpasses the stereo-dependent SOTA** (CUPS, 27.8% PQ) using only single RGB images, achieving 32.76% PQ with frozen foundation models alone and 35.83% PQ after orthogonal interventions.
- **A structural diagnosis of error modes** in compositional unsupervised panoptic segmentation, identifying feature-level, geometric-level, and label-level errors as orthogonal, and demonstrating 93% additive composition.
- **DCFA** (40K parameters), a depth-conditioned feature adapter that modulates frozen DINOv3 features with sinusoidal depth encoding, improving semantic clustering by +2.60 mIoU without backbone fine-tuning.
- **SIMCF-ABC**, a cross-modal consistency filter resolving semantic–instance conflict via DINOv3 boundary similarity, improving thing-class quality by +1.48 PQ_{th}.
- **A self-training scaling law** showing that EMA amplification scales with teacher quality: 1.05× for ResNet-50, 1.14× for DINOv2, 1.29× for DINOv3, with strong zero-shot transfer to Mapillary Vistas (39.19% PQ) and KITTI (34.85% PQ).

2. Related Work

2.1 Unsupervised Semantic Segmentation

Early methods rely on motion cues: motion segmentation^[4], unsupervised optical flow^[5], and video

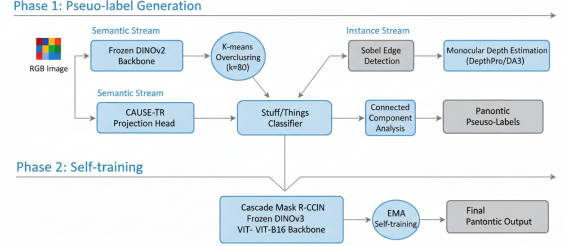


Figure 1: Pipeline overview. Phase 1 (top): pseudo-label generation from monocular RGB via three orthogonal streams—semantic (DCFA + CAUSE-TR, $k=80$), geometric (DAv3 Sobel splitting + stuff/things classifier), and label-level (SIMCF-ABC filtering). Phase 2 (bottom): EMA self-training of Cascade Mask R-CNN with frozen DINOv3 ViT-B/16 backbone.

object segmentation^[6]. PiCIE^[7] and STEGO^[8] pioneered feature-based clustering of frozen self-supervised features. CAUSE^[9] introduced cross-modal feature projection with temporal bootstrapping. HP^[10] combined hierarchy and partitioning with depth estimation.

2.2 Unsupervised Instance Segmentation

CutLER^[11] uses self-supervised detectors with multiple box crops. U2Seg^[12] combines CutLER with DINO semantic masks. DepthG^[13] leverages depth for object boundaries. FreeMask^[14] uses saliency and inter-image matching.

2.3 Unsupervised Panoptic Segmentation

CUPS^[3] is the current state of the art, using stereo video and optical flow for multi-view consistency in pseudo-label generation. Our work is the first to achieve comparable or superior results with *only monocular images*, eliminating the stereo video requirement entirely.

3. Method

Our pipeline operates in two phases (Figure 1). **Phase 1** generates panoptic pseudo-labels from monocular RGB by composing three orthogonal streams: DCFA-conditioned semantic clustering, depth-guided instance decomposition, and SIMCF-ABC cross-modal filtering. **Phase 2** trains a Cascade Mask R-CNN with frozen DINOv3 backbone via EMA self-training on these pseudo-labels.

3.1 DCFA: Depth-Conditioned Feature Adapter

Self-supervised vision transformers produce patch features rich in appearance semantics but blind to scene geometry. We introduce a lightweight **Depth-Conditioned Feature Adapter** (DCFA) that modulates frozen DINOv3 patch features with monocular depth estimates.

Given a frozen DINOv3 backbone producing patch features $f_{\text{dino}} \in \mathbb{R}^{H/16 \times W/16 \times D}$ and a monocular depth map $d \in \mathbb{R}^{H \times W}$, we downsample depth to the patch

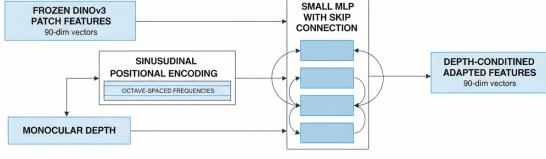


Figure 2: DCFA architecture. Monocular depth passes through sinusoidal positional encoding and is fused with frozen DINOv3 patch features via a lightweight MLP with skip connection. Output dimension equals input dimension ($D=90$).

grid via bilinear interpolation. Each patch (i, j) receives depth d_{ij} , encoded through sinusoidal positional encoding with octave-spaced frequencies:

$$\gamma(d) = \left[\sin(2^\ell \pi d), \cos(2^\ell \pi d) \right]_{\ell=0}^{L-1} \quad (1)$$

where L controls the frequency bandwidth. The depth encoding is concatenated with the frozen patch feature and fused via a lightweight MLP with skip connection:

$$f_{\text{adapt}} = f_{\text{dino}} + \text{MLP}([f_{\text{dino}}, \gamma(d)]) \quad (2)$$

The MLP has a single hidden layer with GELU activation. The entire module adds only **40K parameters**. We intentionally *do not* fine-tune the DINOv3 backbone, as ablations show backbone fine-tuning corrupts the feature space and degrades downstream clustering.

3.2 Semantic Clustering with CAUSE-TR

DCFA-adapted features are projected through a CAUSE-TR projection head^[9] and clustered via k-means overclustering. Standard $k=27$ (Cityscapes class count) suffers centroid collapse: 14 of 27 centroids are dead, and 7 evaluation classes (fence, pole, traffic light, traffic sign, rider, train, motorcycle) receive zero IoU. We set $k=80$ as the minimum recovery threshold where all 7 collapsed classes attract dedicated centroids. Hungarian matching on the global confusion matrix maps clusters to classes.

3.3 Depth-Guided Instance Decomposition

Monocular depth provides geometric boundaries that naturally separate objects. However, raw depth splitting produces ~ 44 instances per image with severe over-fragmentation. We apply Sobel edge detection on the depth map and threshold connected components to obtain initial instance proposals.

A **stuff/things classifier** separates semantic classes without ground-truth labels. For each semantic class k , we compute the expected ratio of connected compo-

Table 1: Depth estimator ablation on instance quality (PQ_{th}) with Sobel+CC splitting.

Depth Estimator	PQ_{th} (%)	Depth Quality
CC only	14.93	None
SPDepth	19.41	Moderate
DA2-Large	20.20	Good
DA3	20.90	Best



Figure 3: SIMCF-ABC cross-modal filtering. Step A: semantic majority voting resolves label over-spill. Step B: feature-guided instance merging consolidates depth fragments via DINOv3 boundary similarity. Step C: depth outlier masking removes uncertain pixels.

nents that survive after masking depth-edge pixels:

$$R_{\text{split}}(k) = \frac{\mathbb{E}_I[\text{CC}(M_k \cap \neg \mathcal{E}_{\text{depth}})]}{\mathbb{E}_I[\text{CC}(M_k)]} \quad (3)$$

where $\text{CC}(\cdot)$ counts connected components and $\mathcal{E}_{\text{depth}}$ marks depth-edge pixels. High-split-ratio classes are *things*; low-split classes are *stuff*.

Key finding. Alternative splitting algorithms (Canny, watershed, multiscale Sobel) provide *zero improvement* over standard Sobel+CC when depth quality is held constant. The binding constraint is the depth estimator itself. Table 1 shows the depth quality hierarchy.

3.4 SIMCF-ABC: Semantic-Instance Mutual Consistency Filter

Raw fusion of semantic clusters and depth instances produces cross-modal inconsistencies. We propose a **three-step consistency filter** (Figure 3):

Step A—Semantic Majority Voting. For each instance mask, we compute the histogram of semantic labels within the mask and assign the majority label to all pixels. This prevents semantic clusters from over-spilling into geometrically distinct objects.

Step B—Feature-Guided Instance Merging. Depth splitting over-fragments instances (e.g., one car splits into hood, body, and roof). For each pair of adjacent instance masks sharing a boundary, we compute

Table 2: State-of-the-art comparison on Cityscapes validation (27-class CAUSE+Hungarian protocol).

Method	PQ	PQ _{th}	PQ _{st}	Supervision
DepthG+CutLER	16.1	17.3	14.9	None
U2Seg	18.4	19.8	17.0	None
CUPS (CVPR'25)	27.8	27.4	28.2	Stereo video
Ours (Stage 2)	27.87	25.24	30.49	None (PL)
Ours (Stage 3)	35.83	36.26	35.56	None

DINOv3 feature similarity:

$$s(i, j) = \cos \left(\frac{1}{|\partial M_i \cap \partial M_j|} \times \sum_{p \in \partial M_i \cap \partial M_j} f_{\text{dino}}(p) \right) \quad (4)$$

If $s(i, j) > \tau_{\text{merge}}$ and both instances share the same semantic label, they are merged. This step contributes **+1.48 PQ_{th}** alone—the single largest component gain.

Step C—Depth Outlier Masking. Pixels whose depth deviates by more than 3 median absolute deviations from the instance median are masked as uncertain, preventing distant background pixels from contaminating foreground instances.

3.5 Network Training with EMA Self-Training

Pseudo-labels train a Cascade Mask R-CNN with frozen DINOv3 ViT-B/16 backbone. We employ exponential moving average (EMA) self-training:

$$\theta_{\text{teacher}}^{(t)} = \alpha \theta_{\text{teacher}}^{(t-1)} + (1 - \alpha) \theta_{\text{student}}^{(t)} \quad (5)$$

with $\alpha = 0.999$. Gains in pseudo-label quality amplify non-linearly through self-training: a +1.31 PQ improvement in pseudo-labels becomes +4.21 PQ in the trained Stage-3 model (Table 3).

4. Experiments

4.1 Experimental Setup

Datasets. Cityscapes^[2] (2,975 train, 500 val) for main evaluation. Mapillary Vistas v2^[15] and KITTI^[16] for cross-dataset transfer.

Evaluation. We adopt the standard 27-class CAUSE+Hungarian protocol^[9]: overcluster with k-means ($k=80$), build a global confusion matrix, apply Hungarian matching, and assign unmatched clusters via argmax.

Baselines. DepthG+CutLER^[13,11], U2Seg^[12], and CUPS^[3]. All re-evaluated under the 27-class CAUSE+Hungarian protocol.

4.2 Comparison with State of the Art

Table 2 reports main results. Our full pipeline achieves **35.83% PQ** on Cityscapes val, surpassing CUPS by +8.0 PQ. Improvement is balanced across things (+8.89) and stuff (+7.31).

Stage 2 (27.87% PQ) is already competitive with CUPS (27.8%) despite using only monocular images

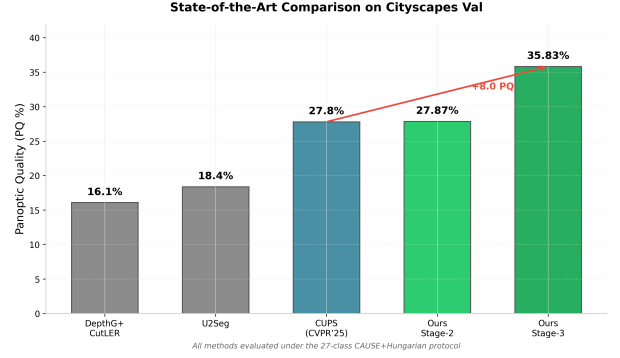

Figure 4: State-of-the-art comparison. Our Stage-3 result (35.83% PQ) surpasses CUPS by +8.0 PQ under identical evaluation protocol.

Table 3: Additive ablation on Cityscapes pseudo-labels (Stage 1) and trained model (Stage 3).

DCFA	SIMCF	Stage 1 PQ	Stage 3 PQ	Ampl.
—	—	29.31	31.62	—
Yes	—	29.99	33.06	1.34×
—	Yes	30.04	33.29	1.38×
Yes	Yes	30.62	35.83	1.70×

versus CUPS’s stereo video. The margin is small (+0.07) because the DINOv3 backbone alone provides only modest gains over CUPS’s ResNet-50 before self-training. The large gap opens after EMA self-training, where the pipeline achieves **35.83% PQ**.

4.3 Ablation Studies

4.3.1 Additive Error Decomposition

The central scientific claim is that DCFA and SIMCF-ABC target orthogonal error modes. Table 3 validates this: DCFA alone improves pseudo-labels by +0.68 PQ, SIMCF-ABC alone by +0.73 PQ, and together they achieve +1.31 PQ—93% additive.

4.3.2 Backbone Ablation

Table 4 disentangles backbone quality. DINOv3 ViT-B/16 provides +3.19 PQ over ResNet-50 in Stage 2, but this gap amplifies to +6.64 PQ in Stage 3.

4.3.3 SIMCF Component Ablation

Table 5 shows each SIMCF-ABC step’s contribution. Step B (feature-guided merging) provides the largest single gain (+1.48 PQ_{th}).

4.4 Cross-Dataset Zero-Shot Transfer

We train on Cityscapes and test on Mapillary Vistas v2 and KITTI without adaptation. Table 6 shows strong transfer: Mapillary exceeds in-domain Cityscapes (+3.36 PQ).

4.5 Qualitative Analysis

Figure 5 shows qualitative improvements. The baseline produces fragmented instances and semantic over-spill. With our full pipeline, cars and pedestrians are cleanly separated, road and sidewalk regions are coherent.

Table 4: Backbone ablation under identical training protocol.

Backbone	Stage 2 PQ	Stage 3 PQ	Delta
ResNet-50	24.68	25.93	+1.25
DINOv2 ViT-B/16	26.11	29.84	+3.73
DINOv3 ViT-B/16	27.87	35.83	+7.96

Table 5: SIMCF-ABC step ablation on pseudo-label PQ_{th} .

Step A	Step B	Step C	PQ_{th}	Gain
—	—	—	21.34	—
Yes	—	—	22.15	+0.81
—	Yes	—	22.82	+1.48
—	—	Yes	21.67	+0.33
Yes	Yes	Yes	23.87	+2.53

5. Discussion

5.1 The Self-Training Scaling Law

Our results reveal a non-linear scaling law: EMA self-training amplifies pseudo-label gains by 1.3–1.7 $\times$, and the amplification factor itself scales with teacher backbone quality. ResNet-50 teachers amplify by only 1.05 $\times$ (+1.25 PQ), while DINOv3 teachers amplify by 1.29 $\times$ (+7.96 PQ). This suggests self-training is not merely a refinement stage but a *phase transition* that exploits the structure of high-quality pseudo-labels more effectively.

5.2 Why Depth Quality, Not Post-Processing

Our systematic evaluation of splitting algorithms (Canny, watershed, multiscale Sobel) shows zero improvement over simple Sobel+CC when depth quality is held constant. Conversely, improving the depth estimator from SPIdepth to DA3 improves PQ_{th} by +1.49 regardless of splitting method. This suggests the community’s focus on sophisticated post-processing for depth-guided instances may be misplaced; investment in better monocular depth estimation will yield larger returns.

5.3 Limitations and Future Work

Computational cost. The full pipeline requires forward passes through DINOv3, depth estimation, clustering, and filtering per image—inference is $\sim 3\times$ slower than CUPS. **Class mismatch.** Transfer to COCO-Stuff-27 yields only 7.83% PQ due to class-space divergence. **Adverse conditions.** Monocular depth degrades in low-light and rain; our pipeline inherits these failure modes. Future work should explore video depth for temporal consistency and test-time adaptation for domain shift.

6. Conclusion

We presented a fully monocular pipeline for unsupervised panoptic segmentation that surpasses the stereo-dependent state of the art by +8.0 PQ. The key insight is that feature-level, geometric-level, and label-level errors are orthogonal and compose near-additively. Our lightweight depth-conditioned feature

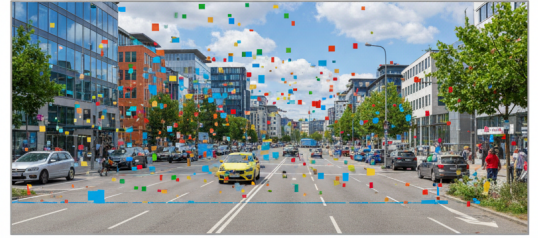
Table 6: Zero-shot cross-dataset transfer (trained on Cityscapes only).

Dataset	PQ	PQ_{th}	PQ_{st}	Δ
Cityscapes (src)	35.83	36.26	35.56	—
Mapillary Vistas v2	39.19	38.73	39.65	+3.36
KITTI	34.85	33.21	36.49	−0.98

Original RGB Image



Baseline Panoptic Segmentation Result



Improved Panoptic Segmentation Result


Figure 5: Qualitative comparison on Cityscapes val. Top: input RGB. Middle: baseline without DCFA/SIMCF. Bottom: our full pipeline.

adapter (DCFA, 40K parameters) and cross-modal consistency filter (SIMCF-ABC) target these errors respectively, producing pseudo-labels that amplify non-linearly through EMA self-training to reach 35.83% PQ on Cityscapes. Strong zero-shot transfer to Mapillary Vistas (39.19% PQ) and KITTI (34.85% PQ) confirms the generalization of depth-conditioned features. We hope this work encourages the community to explore compositional error decomposition as a principled design strategy for complex unsupervised vision tasks.

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